

Data Analysis using R Programming Lab Questions (III SEM)

Course Code:CS1602-1

YEAR :2023-24

PART A	
1.	Create R program to write a user defined functions and perform the following operations on user input. - Check the number is even or odd - Print squares of numbers in sequence. - Find the factors of number. - Create vector of integers and sort them in ascending/descending order.
2.	Create matrices in R and perform the following operations on them. - Addition, Subtraction of two matrices and product of matrix and its transpose. - Column sum, Mean across rows, Total Sum of a matrix and Sort the matrix elements across columns in ascending order. - Inverse of the matrix. - Find row and column index of maximum and minimum value in a given matrix.
3.	Create R program and perform the following operations on them. - Take user name as input string, display the number of characters present in the string, convert the string into uppercase and display the middle character of the string. - Create function called is_palindrome() that determines whether or not a given string is a palindrome. The function should take a single parameter. - Create an ordered factor of 20 alternating pairs of 1s and 2s, label the data with the "ON" and "OFF".
4.	Write a R program to create a list containing a vector, a matrix and a list and give names to the elements in the list. - Count number of objects in a given list and Find the length of the first vector of a given list. - Access the first and third element of the list and extract all elements of vector except the third element vector. - Add element at the end of the list and Update the third element of the list and remove the last element.

	d. Merge list of vectors to original list, convert a second and last element of a given list into vector and add these two vectors.
5.	Consider the data, <code>X <- c(21, 62, 10, 53)</code> and Labels <code><- c("London", "New York", "Singapore", "Mumbai")</code>. Write the R program to create the following Pie chart - Create a simple Pie chart with Labels that describes the slices. - Create a chart with rainbow colors and add an appropriate title. - Create a 3-D chart with legend and slice percentages and save that chart
6.	Write the R program to plot the following charts - Create the Bar chart for <code>H <- c(7,12,28,3,41)</code> give the chart file a name and plot the chart. - Plot the line chart for the data in (a) and save that chart - Consider the data <code>v <- c(19, 23, 11, 5, 16, 21, 32,14, 19, 27, 39)</code>. Create a simple histogram for Range of X and Y values. - Create a histogram using non-uniform width for the vector v.
7.	Consider the dataset <code>mtcars</code> in R and perform the following. - Write a R program to create the Box plot with chart file name - Create a simple scatterplot to Plot the chart for cars with weight between 2.5 to 5 and mileage between 15 and 30. - Create matrix of scatterplots for the relation between <code>wt(weight)</code>, <code>mpg(miles per gallon)</code>, <code>disp(displacement)</code> and <code>cyl(number of cylinders)</code>. Save the chart.
8.	Create R program and perform the following operations on them. - Create a vector <code>x <- c(12,7,3,4.2,18,2,54,-21,8,-5)</code> and find mean . - Apply trim option and NA option. - Find the median for the vector <code>v <- c(2,1,2,3,1,2,3,4,1,5,5,3,2,3)</code> and calculate the mode for the character vector <code>charv <- c("o","it","the","it","it")</code> using the user function.
	PART B
1.	Create a Data Frame in R for maintaining Employee Details using .csv file that contains 5 Employee Details consisting of Employee ID, Employee name, Salary, and Joining date and perform the following analysis on it. - Get the structure of the data frame. - Add the Department column to existing data frame using <code>cbind()</code> function and add 2 employees details <code>rbind()</code> function. - Get the employee's name , salary details and Get the maximum salary. Get the persons in IT department whose salary is greater than 600 and get the employee name

	who joined on or after 2014
2.	Perform the following operations on Data Frame - Create the first data frame that contains student Details consisting of Student ID, and marks for three subjects, namely, English, Maths, and Science. Create the second data frame with column Student ID and State. Join both the data frames on Student ID. - Add the new subject 'Social' to joined data frame of (a) using cbind() function and add 2 more student details using rbind() function. Get the Student ID and state details. Get the student details belongs to "Karnataka" State and Maths marks greater than 50. - Use the data frame of (b) and organize the data with respect to Student ID and State using the melt() function. Use the molten data and reshape the data to its original form using cast() function.
3.	Write an R program that accomplishes the following tasks by implementing user- defined functions: - Define a function to calculate the age of a person given their birthdate. The function should take the birthdate as input and return the age in years. - Create a function that calculates the number of days between two given dates. The function should accept two dates as inputs and return the difference in days. - Implement a function that takes a date and time in the format "YYYY-MM-DD HH:MM:SS" and converts it into a more user-friendly format, displaying the day of the week, month, day, and year along with the time in "Hour:Minute AM/PM" format. - Create a function that receives a date and time in the format "YYYY-MM-DD HH:MM:SS" and returns the day of the year and the week number of the year.

4.	Create a R data frame to manage Student Scores using a CSV file and perform following analysis tasks: - Write an R function named <code>read_csv_data</code> that takes a file name as an argument and returns a data frame containing the data from the CSV file. Create a function to find and display the student with the highest total score (<code>Math_Score + Science_Score + English_Score</code>) from the loaded data. - Write an R function named <code>calculate_pass_percentage</code> that takes a data frame and a passing threshold as arguments and returns the percentage of students who passed all subjects (scores above the threshold). - Use the <code>dplyr</code> library to write a function named <code>calculate_subject_averages</code> that calculates the average scores for each subject (Math, Science, English) and returns them as a data frame. - Create a function named <code>save_passed_students</code> that takes a data frame and a passing threshold as arguments. The function should save a new CSV file named <code>passed_students.csv</code> containing only the rows of students who passed all subjects with scores above the threshold.
5.	Create the data frame that contains student Details consisting of Student name, and marks for three subjects, namely, Physics, Chemistry and Mathematics. Perform the following analysis on it using <code>dplyr</code> library. - Use <code>Select()</code> function to print only column details starts with "Ph" and print everything that contains "cs" - Display data in which Mathematics marks lie between 90 and 94 using <code>filter()</code> - Display the physics mark on sorted order using <code>arrange()</code> function and rename the Chemistry column to <code>CHEMISTRY</code> using <code>rename()</code> function - Create the resultant data frame that contains new variable <code>TotalMarks</code> using <code>mutate()</code> function.
6.	Write a R program and perform following operations using tidyverse packages - use the <code>map()</code> function to calculate the square of each value in a vector <code>c(2, 4, 10, 15, 20)</code> and calculate mean value of each vector in <code>list<- list(c(1, 2, 3), c(4, 5, 6), c(7, 8, NA))</code>. - use <code>str_replace()</code> function to replaces a pattern or string with another string and use <code>str_length()</code> to find the length of each element of vector <code>x<- c("R", "programming", "Language")</code>.



	c. create the dataframe with letters and numbers and display the bar graph using ggplot()
7.	Create the data frame that contains revenue details of 4 years consisting of variables namely, Group, Year, Qtr.1 Qtr.2 Qtr.3 and Qtr.4. Perform the following analysis on it using tidyr library. - Create a data frame by Re-structuring the time component Qtr.1 Qtr.2 Qtr.3 and Qtr.4 as an Quarter and Revenue using gather() function. - Use the result obtained from (a), Split the Quarter variable into Time_Interval and Interval_ID using separate(). - Use the result obtained from (b) , re-unite the Time_Interval, Interval_ID variables and re-create the original Quarter variable using unite() function. - Use the result obtained from (c), reshape the long format to wide format using spread() function.
8.	Consider mtcars dataset and perform the following statistical analysis - Calculate the interquartile range, variance and Standard Deviation of "mpg" variable. - Calculate covariance of the mpg and "wt" columns in the mtcars and create a covariance matrix of these two variables. - Calculate the correlation between "mpg" and "wt". - Implement regressor model to establish the relationship between "mpg" as a response variable with "disp","hp" and "wt" as predictor variables. From the model,get the Coefficients and Create Equation for Regression Model and Apply Equation for predicting mileage for a car with disp = 221, hp = 102 and wt = 2.91



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